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DISABILITY AND REHABILITATION: ASSISTIVE TECHNOLOGY
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REVIEW

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Implementation of welfare technology: a state-of-the-art review of knowledge gaps and research needs

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ABSTRACT

Purpose: To offer guidance for future welfare technology research, this review provides an overview of current knowledge gaps and research needs as reported in primary scientific studies addressing the implementation of welfare technology for older people, people with disabilities and informal caregivers.

Materials and methods: This paper conducted a state-of-the-art review based on systematic searches in 11 databases followed by a descriptive qualitative analysis of 21 selected articles.

Results: Knowledge gaps and research needs were identified concerning two categories: research designs and populations and focus of research. The articles reported needs for comparative studies, longitudinal studies, and demonstration trials as well as the development of co-design processes involving technology users. They also called for studies applying a social system theory approach, involving healthy and frail older adults, representative samples of users within and across countries, informal and formal caregivers, inter-and multidisciplinary teams, and care organizations. Moreover, there are reported needs for studies of acquirement, adoption and acceptance of welfare technology, attitudes, beliefs, and context related to welfare technology, caregiver perspectives on welfare technology, services to provide welfare technology and welfare technology itself.

Conclusions: There are considerable knowledge gaps and research needs concerning the implementation of welfare technology. They relate not only to the research focus but also to research designs, a social system theory approach and study populations.

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Ambient assisted living; digital assistive technology; implementation; knowledge gaps; research needs; telecare; welfare technology

➤ IMPLICATIONS FOR REHABILITATION

- When planning for the implementation of welfare technology for older people and persons with disabilities, it is important to be aware that necessary evidence and guidance may not always be available in peer-reviewed scientific literature but considerable knowledge gaps and research needs remain.
- Actors implementing welfare technology are encouraged to include researchers in their projects to study, document and report experiences made, and thereby contribute to building the evidence base and supporting evidence-based implementation.

Introduction

There is an increasing interest in utilizing new digital technologies, welfare technology, to address the double demographic challenge of a growing proportion of older people and decreasing proportion of people of working age. Despite its potential to improve the quality of life of people with functional limitations of all ages and to reduce societal costs, the large-scale implementation of welfare technology has been slow and narrow in scope [1–5]. Reported implementation barriers include high costs, lack of competence, lack of knowledge about available solutions, differences between sectors, technical problems, legal complications, resistance to change among personnel, limited research, and weak analyses of needs. Further, identified challenges for implementation relate to financing, organizational issues, communication and cooperation across stakeholders, evaluation, IT infrastructure, service, support, logistics and reacquisition [6–9].

There is no official definition for welfare technology, but its purpose, according to the Swedish National Board of Health and Welfare, is to maintain or increase safety, activity, participation, or independence for individuals with disabilities or those who are at risk for disability [10]. The concept welfare technology is newer than the phenomenon of technology use in healthcare and social care, and there are overlapping concepts that are used differently, for example, eHealth, teleHealth, telecare, and telemedicine. However, there are limitations of the above definition of welfare technology. As pointed out by Frennert [11], one limitation is that the definition does not include social and organizational aspects,

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structures and how healthcare and social care practices are affected by welfare technology. Frennert and Baudin [12] have also explored the welfare technology concept in Swedish municipal elderly care. They found that it was recognized as an enabler, a simplifier and as a support for both patients and professional care givers, which corresponds to the definition of welfare technology.

Recently, the World Health Organization (WHO) embraced the normalization of technology as a part of everyday life by launching a new initiative: Digital and Assistive Technologies for Ageing (DATA). The DATA initiative "seeks to encourage the development, synthesis, and use of solutions that promote access to affordable, quality, digital and assistive technologies for people with impairment or decline in physical or mental capacity, with a particular focus on older people" [13]. The digital technology component in the WHO DATA initiative resembles what is defined as welfare technology by the National Board of Health and Welfare in Sweden, i.e., digital technology that aims at maintaining and increasing safety, activity, participation, or independence for a person who has or is at an increased risk of impairment [10]. The Swedish definition of welfare technology is narrower than the broad Nordic description of welfare technology as technology that improves the lives of those who need it [1].

Using the Swedish definition of welfare technology, the purpose of this review is to offer guidance for future welfare technology research and thereby contribute to mitigating barriers and challenges related to lack of knowledge regarding the implementation of welfare technology. Its objective is therefore to provide an overview of current knowledge gaps and research needs as reported in primary scientific studies that addresses the implementation of welfare technology for older people, people with disabilities and informal caregivers.

Table 1. Number of identified records by database.

Database	Number of identified records
Academic Search Elite (EBSCOhost)	208
APA PsycINFO (EBSCOHost)	35
ASSIA (ProQuest)	199
Cinahl Plus (EBSCOHost)	63
IBSS (ProQuest)	15
PubMed (https://pubmed.ncbi.nlm.nih.gov)	660
Scopus (www.scopus.com)	1,367
SocINDEX (EBSCOHost)	6
Social Services Abstracts & Sociological Abstracts (ProQuest)	14
Web of Science Core Collection (www.webofknowledge.com)	1,484
Total	4,051

Method

This study conducted a state-of-the-art review with a focus on highlighting areas in need of further research [14]. Applicable methodological principles of the Cochrane Handbook for Systematic Reviews of Interventions [15] were followed, particularly regarding the search strategy, which involved several databases and the use of both free-text and controlled vocabulary. An *a priori* protocol that includes this review was registered on PROSPERO (CRD42020170051).

Data sources and searches

Eleven electronic databases (see Table 1) were searched on 16 March 2020, by an academic librarian (author SLS) using a comprehensive search strategy that was developed by all authors. Divided into three categories, the search terms included both free-text and controlled vocabulary. The first category related to population: older persons, people with disabilities and informal caregivers (#1–3 in Table 2). The second category focussed on intervention (#5 in Table 2): welfare technology and related terms, such as smart home and environmental control. The third category concerned outcomes in terms of implementation (#6 in Table 2), i.e., activities designed to put something into practice. The Boolean operator AND was used to combine the three primary categories of search terms. To exclude studies regarding children, terms relating to children and youths were omitted using the Boolean operator NOT (#4, 7–9 in Table 2).

As the term welfare technology was introduced in 2007, the searches were limited to peer-reviewed material published from 1 January 2007 until the date of the search. In the databases, the Web of Science and Scopus, further limits regarding subject areas were applied, and document types were limited to articles, reviews and conference or proceeding papers. The reason for this was to reduce records of technically focussed research where implementation has a different meaning. In the Web of Science,

Table 2. Search strategy used in Scopus (limitations are specific to this database).

#1	TITLE-ABS-KEY ("aged people*" OR "aged person*" OR "aged women*" OR "aged woman" OR "aged men*" OR "aged man" OR "aged patient*" OR ageing OR ageing OR elder* OR "old* people*" OR "old* patient*" OR "old* age*" OR "old* person*" OR "old* women*" OR "old* woman" OR "old men*" OR "old man" OR senior* OR "housing for the elderly" OR "nursing homes" OR "assisted living facilit*")
#2	TITLE-ABS-KEY (family OR families OR relative OR relatives OR "next of kin*" OR "significant other*" OR caregiver* OR "care giver*" OR care-giver* OR carer* OR spouse* OR partner OR partners OR husband OR husbands OR wife* OR wive*)
#3	TITLE-ABS-KEY (disab* OR impair* OR disorder* OR retard* OR handicap* OR (physical W/2 deficien*) OR (mental W/2 deficien*) OR deaf* OR blind* OR "hard of hearing*" OR "functional limitation*" OR "functional problem*" OR "functional deviation*" OR "physically challenged" OR "visual loss")
#4	#1 OR #2 OR #3
#5	TITLE-ABS-KEY ("intelligent assistive" OR "digital assistive" OR "welfare technol*" OR gerontech* OR gerotech* OR "smart home*" OR "Internet of Things" OR iot OR alarm OR alarms OR "environmental control*" OR "ambient assisted living")
#6	TITLE-ABS-KEY (implement* OR apply* OR applie* OR adopt* OR introduc*)
#7	#4 AND #5 AND #6
#8	TITLE-ABS-KEY (child* OR offspring* OR adolescen* OR teen* OR youngster* OR "young people*" OR youth* OR student* OR pupil* OR puberty OR minor* OR underage* OR "under age*" OR juvenile* OR school-age* OR boy* OR girl* OR "young adult*" OR "early adult*" OR pediatric* OR paediatric* OR infant* OR toddler*)
#9	#7 NOT #8 Limitations: (LIMIT-TO (PUBYEAR, 2020) OR LIMIT-TO (PUBYEAR, 2019) OR LIMIT-TO (PUBYEAR, 2018) OR LIMIT-TO (PUBYEAR, 2017) OR LIMIT-TO (PUBYEAR, 2016) OR LIMIT-TO (PUBYEAR, 2015) OR LIMIT-TO (PUBYEAR, 2014) OR LIMIT-TO (PUBYEAR, 2013) OR LIMIT-TO (PUBYEAR, 2012) OR LIMIT-TO (PUBYEAR, 2011) OR LIMIT-TO (PUBYEAR, 2010) OR LIMIT-TO (PUBYEAR, 2009) OR LIMIT-TO (PUBYEAR, 2008) OR LIMIT-TO (PUBYEAR, 2007)) AND (LIMIT-TO (DOCTYPE, "cp") OR LIMIT-TO (DOCTYPE, "ar") OR LIMIT-TO (DOCTYPE, "re")) AND (LIMIT-TO (SUBJAREA, "MEDI") OR LIMIT-TO (SUBJAREA, "SOCI") OR LIMIT-TO (SUBJAREA, "DECI") OR LIMIT-TO (SUBJAREA, "ENVI") OR LIMIT-TO (SUBJAREA, "HEAL") OR LIMIT-TO (SUBJAREA, "BUSI") OR LIMIT-TO (SUBJAREA, "NURS") OR LIMIT-TO (SUBJAREA, "PSYC") OR LIMIT-TO (SUBJAREA, "MULT") OR LIMIT-TO (SUBJAREA, "ECON"))

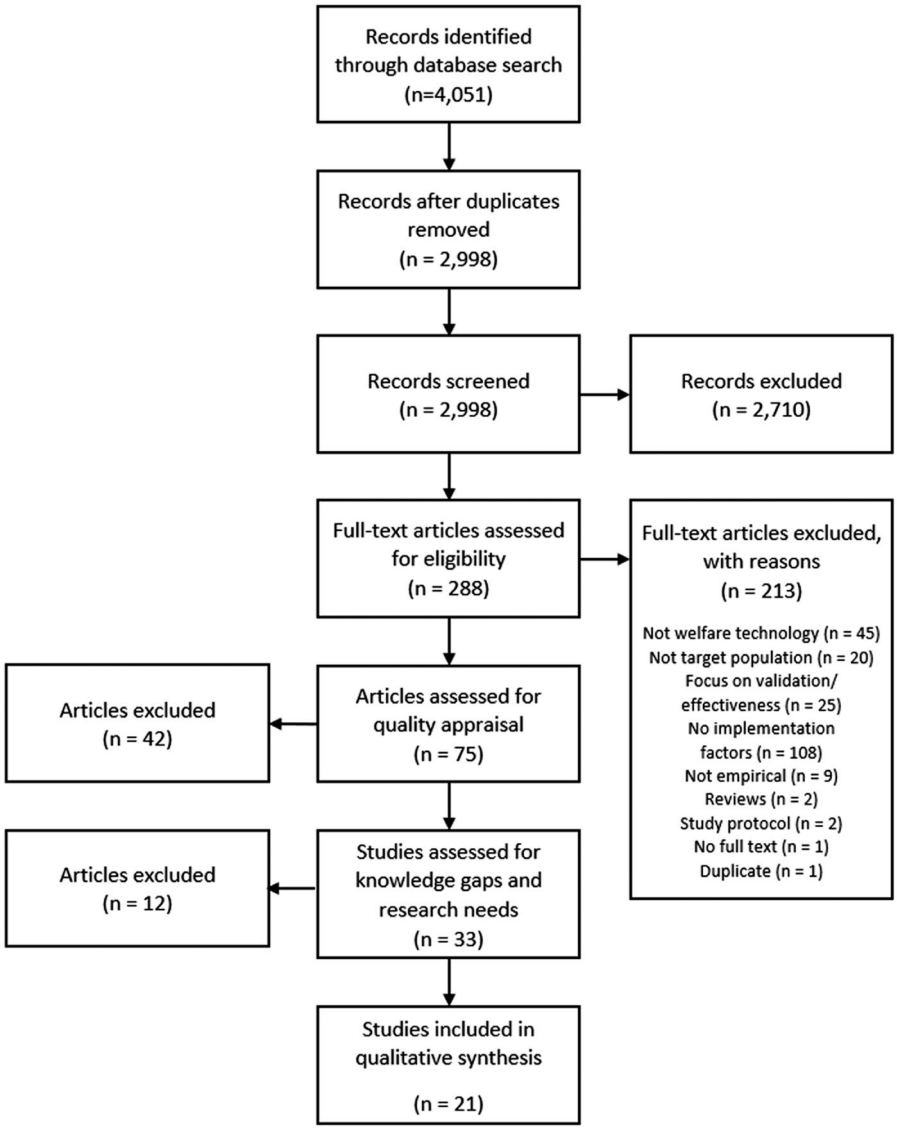


Figure 1. Expanded PRISMA flow diagram.

early access documents were included. No language or study design restrictions were applied to the search.

Study selection

A summary of the study selection process is shown in an expanded PRISMA flowchart in Figure 1. The database searches identified a total of 4,051 records for potential inclusion. These records were downloaded to the reference management programme EndNote, and duplicates were removed by an academic librarian (author SLS) by applying Bramer's [16] de-duplication method, which resulted in the exclusion of 1,053 duplicates. The titles and abstracts of the remaining 2,988 titles were screened against the inclusion criteria.

This review included quantitative and qualitative empirical studies that assessed the implementation of welfare technology from the perspectives of older people, people with disabilities or informal caregivers and had been reported in peer-reviewed articles and published in scientific journals or conference proceedings. Technical reports limited to describing or evaluating features of welfare technology were excluded.

Authors JB and VZ screened the titles and abstracts, and full-text assessment was undertaken by authors JB, CG and VZ. Following full-text assessment, the quality of the articles was appraised by authors JB, CG and VZ using the MMAT mixed methods appraisal tool [17]. MMAT was designed to appraise the methodological quality of five categories of studies: randomized controlled trials, non-randomized studies, quantitative descriptive studies, qualitative studies, and mixed methods studies. The appraisal of the studies resulted in a rating of methodological quality from 0 to 5, where studies rated 3 to 5 were included ($N = 33$).

Any disagreements between the three researchers during this process were discussed until consensus was reached. Reasons for exclusion were recorded.

Data extraction

A data extraction form was developed that included the aim and design of the study, sample characteristics, type of welfare technology, barriers to and facilitators of implementation and reported knowledge gaps and research needs. Details about the articles were also extracted, including publication year and country of study.

Table 3. Example of coding and categorization.

Main category	Preliminary categories	Codes
Research designs and populations	Research design	Compare different groups
		Systematic literature reviews
		Perform demonstration trials
		Develop a co-design process
		Explore longitudinal mechanisms
		Regions and countries
		Different countries
		Interdisciplinary teams
		Compare different groups of informants
		Focus on the patient and families
	Setting and sample	Equally sampled age groups
		Integrate perspectives of older adults' spouses and relatives
		Capture perspectives of a diverse group of health care workers

Among the 33 included articles, 21 reported knowledge gaps and research needs. Thus, the analysis in this review is based on knowledge gaps and research needs reported in these 21 articles.

Synthesis

A qualitative descriptive analysis [18] approach was conducted. The first stage was line-by-line coding, as each sentence was read to identify underpinning themes about knowledge gaps and research needs as reported in primary scientific studies addressing the implementation of welfare technology for older people, people with disabilities and informal caregivers. These were labelled with a code. The identified codes were then organized into preliminary categories. This stage was conducted by VZ and discussed among JB, CG and VZ. They interpretated, abstracted and conceptualized the preliminary categories into two main categories, of which one consisted of five sub-categories. An example of the coding and categorization is given in Table 3.

Results

The aims or research questions of the included studies, and the knowledge gaps and research needs they report are compiled in Table 4. The applied method resulted in two emerging descriptive categories of knowledge gaps and research needs relating to the implementation of welfare technology for older people, people with disabilities, and informal caregivers. The main categories are: (i) research designs and populations, and (ii) focus of research.

Research designs and populations

The first main category explores a broad research gap regarding research designs, covering requests for longitudinal, larger, comparative, and international studies, as well as smaller studies comparing different groups, demonstration trials and a focus on patients and families. Further settings and samples are called for to increase diversity and include equality integrated perspectives. In terms of research designs and populations, the included articles expressed the need for systematic literature reviews of socio-technological challenges from the perspectives of community, organization, and individuals [34]. They also called for demonstration trials with strategies responsive to multi-implementation barriers [36] and the development of co-design processes involving users [31]. Comparisons of opinions and attitudes among older adults who are healthy, and frail were also requested [33]. The articles reported the need for studies with samples within and across countries, cultures, and health-care systems [25,27]. Studies of informal caregivers [24], spouses and relatives [25], and patients and families as end users of welfare technology [3] across age groups [30] were called for. Furthermore, there were reported needs for research on care organization [25], diverse groups of healthcare workers [36] and inter- and multidisciplinary teams representing computer science and engineering, psychology, neuroscience, social work, nurses, physical therapy, speech therapy, physicians, and health information technology experts [21]. In addition to the research designs and methods mentioned above, the reviewed literature expressed a need for systematic literature reviews [34], demonstration trials [36], comparative studies [24,33], and longitudinal studies [22,23].

Focus of research

The included articles called for studies of specific phenomena. The identified knowledge gaps and research needs were categorized into five interlinked and partly overlapping sub-categories. These sub-categories include: (i) acquirement, adoption and acceptance; (ii) attitudes, beliefs and context; (iii) caregiver perspectives; (iv) services; and (v) technology.

The main category covers a broad variety of research perspectives. The research gap includes individual perspectives, including experiences of patients, relatives, and staff, organizational (care providers') perspectives, for example acquirement, services, and organizational infrastructure issues, as well as a societal perspective on challenges in healthcare systems and technology related barriers. This is interpreted as a call for a social system theory approach with micro-, meso- and macro-level perspectives on research about implementation of welfare technology.

Acquirement, adoption and acceptance

The articles report a need for research about mechanisms that impede acquiring welfare technology. This includes why acquirement processes are aborted or not started, and how cultural aspects and the organization of the local and national healthcare system influence technology acquirement [33]. There is also a need for a deeper understanding of factors that influence perceptions of need [19] and differences in users' needs between countries [27].

Other knowledge gaps and research needs include drivers and barriers towards acceptance [24] and differences in acceptance between countries, healthcare systems and cultures [27]. Moreover, there is a need for studies on potential facilitators to promote adoption [28], including best practices for practitioners [35], and emotional involvement in the adoption process [21,25]. There is also a need to study how older adults facing challenges in independent living evaluate and decide between various options [23]. Similarly, there is a need for studies on barriers to

Table 4. Aim or research questions of included studies and reported knowledge gaps and research needs.

Author, year (ref)	Country	Study aim or research questions (Quotes)	Knowledge gaps or research needs (Quotes)
Arthanat et al., 2019 [35]	USA	1. To what extent has smart home technology been adopted by older adults? What types of smart home devices have high rates of adoption, potential for adoption, and rejection among the population? 2. What are the demographic and health-related factors related to the overall adoption profiles of smart home technology by older adults and to what extent do the factors contribute to their smart home ownership and readiness?	Future research may examine the explored factors in an intervention context with smaller cohorts of clients to further elucidate barriers and best practices for practitioners to promote SH [smart home] technology adoption.
Bentley et al., 2014 [37]	UK	To examine the reasons why people choose not to adopt Telecare when it may be an appropriate intervention in their circumstances	An ageing population and increased life expectancy have led to a focus on empowering older people to lead fulfilling and autonomous lives, which may alter the demand for products and services within older age groups. Our findings are consistent with those of other research projects and add to the call for transformation of the way Telecare is designed, marketed and delivered to older people. People need to make their own choices about which technologies are useful and relevant to them, which should be facilitated through the provision of sophisticated, safe and personalisable options (e.g., Telecare *via* smartphone apps) rather than generically providing people with stigmatizing pendant alarms.
Bäccman et al., 2020 [22]	Sweden	To explore and understand users' expectations and experiences in a care situation where digital assistive technology was applied. In this case, the shower situation	Therefore, future studies should attempt to observe the effects of a more frequent use, allowing the users to learn more about the robotic shower's weaknesses and strengths. In addition, as the municipality have found positive ripple effects on other behaviours, future studies should also expand their focus to include how the use of DATs can affect other types of behaviours. In particular, future research should include the important temporal effects (immediate value-creation and long-term) described in this research.
Cajita et al., 2018 [28]	USA	To assess the perceptions of older adults with heart failure regarding the use of mobile technology and to identify potential of and barriers to mHealth adoption	Future researchers who seek to implement mHealth based interventions should address person-related, technology-related, and contextual barriers, and simultaneously capitalize on the influence of potential facilitators, such as a physician's recommendation, to promote mHealth adoption.
Courtney, 2008 [19]	USA	To explore the meaning of privacy and how perceptions of privacy by older adults living in residential care facilities might affect their willingness to adopt smart home information technologies	Further exploration of the factors influencing older adults' perceptions of smart home IT need is necessary.
Fritz et al., 2016 [21]	USA	Through the lens of socially constructed values and beliefs, the purpose of this research was to explore older adults' perceptions and descriptions of smart home monitoring, and empower older adults by encouraging them to share their views	Continued exploration regarding the role of culture in the adoption of smart home monitoring is needed. Understanding the role of culture in adoption practices may inform design and marketing, and may elicit a focus on ethical principles of autonomy and the right to self-determination within the healthcare delivery system. Health-assistive smart home research should continue to be interdisciplinary. Ageing is a complex process that holistically involves mind, body, and spirit. No single discipline can holistically address the journey through ageing. Key interdisciplinary team members currently include computer science and engineering, psychology, neuroscience, social work, and nurses. Future interdisciplinary teams should consider the inclusion of physical therapy, speech therapy, physicians, and health information technology experts. Integration of information into the electronic medical record will be important as personal health data becomes integrated at a national level on the new healthcare information cyber highway. Further concepts that require exploration are trust, caregiver perspectives, features and functionality, human factors engineering, emotion, and cost.
Jaschinski and Allouch, 2019 [24]	Netherlands	To explore the perceptions and needs of informal caregivers with regard to ambient assisted living technologies and examine if and how these perceptions differ from the care receivers' perceptions	Despite its limitations, this study offers several interesting avenues for future research. First, given the fact that care tasks and the (subjective) task pressure can be highly diverse, future studies should compare different groups of informal caregivers [see

(*continued*)

Table 4. Continued.

Author, year (ref)	Country	Study aim or research questions (Quotes)	Knowledge gaps or research needs (Quotes)
			caregiver roles by Consolvo et al. (2004) and other contextual factors: culture, age, personal innovativeness] and explore how these factors affect the perception of AAL technologies. Second, the two male informal caregivers in our studies were fairly positive towards AAL technologies. Although informal caregivers are typical female, it would be interesting to focus more on the male perspective in future research. Third, we believe that our in-depths findings can be leveraged for quantitative approaches to further investigate the influence of the identified drivers and barriers towards AAL acceptance. Fourth, considering that our results describe perceptions in a preadoption phase, it would be interesting to investigate if and how the meaning of the found drivers and barriers change in a later acceptance stage when technologies are integrated in the everyday care practices.
Kolkowska et al., 2016 [34]	Sweden	To investigate the socio-technical challenges in implementation of monitoring technologies in elderly care	… the framework and the categories need to be further verified and refined. The first step in our future research will be conducting systematic literature reviews focussing on socio-technical (ST) challenges in relation to the three levels identified in this study: community, organization and individual, aiming to refine and complement the categories identified and to assess the coherence with previous research.
Nakrem et al., 2018 [20]	Norway	To explore how healthcare professionals have experienced the introduction of digital medicine dispensers in home healthcare services and their influence on patient-caregiver relationships	Ethical implications for both individuals and societies need to be further discussed.
Nilsen et al., 2016 [3]	Norway	The aim of this study was to explore resistance to implementation of welfare technology in five municipalities in Norway	In studies as the one at hand, many factors influence the context. In order to reduce complexity, we have omitted several factors. Central and important stakeholders like the patients and next of kin have not been included in the study. This is because we wanted to focus on the employees, but at the same time, we recognize the patient and his/her family as the real end user of the welfare technology. Focus on the patient and families will need to be included in future studies.
Okonji and Ogwezzy, 2019 [30]	Nigeria	This study explored barriers to adoption of assistive technologies among visually impaired people in five southern states in Nigeria	Future studies should investigate how the type of vision impairment and visual acuity could determine awareness and decisions to adopt computer assistive technologies. In addition, the list of possible barriers is not exhaustive of the spectrum of barriers to adoption of ACTs. Future research should explore a broader list of barriers. While our sample may be a true reflection of prevalence of vision impairment among the age-groups (i.e., older people are more affected with vision impairment than younger adults because vision impairment is associated with ageing), there are possibilities that the results are skewed to reflect more of the responses of the older adults' age group than the younger. Future research may focus on covering larger and equally sampled age groups.
Olsson et al., 2013 [26]	Sweden	The aim of the present intervention study was to describe and explore the use and experiences of using a passive positioning alarm, over time, in daily life among persons with dementia and their spouses	… future research is needed to explore the experiences of persons with dementia in later stages of the disease. Persons with severe dementia may experience deterioration of their verbal communication skills, which is why a longitudinal observation study should be considered.
Olsson et al., 2016 [38]	Sweden	To describe perceptions of a passive positioning alarm (PPA) among persons with mild dementia	Future development of advanced technology for PwMDs [people with mild dementia] should include potential users' perceptions and experiences of the design prior to implementation. If this is done, we believe interventions will truly live up to their potential to support both PwMDs and their relatives.
Peek et al., 2016 [23]	Netherlands	To explore which factors influence the level of use of various types of technology by older adults who are ageing in place and to describe these factors in a comprehensive model	Additionally, more research is needed to better understand how older adults evaluate and decide between the various (technological) options that are available to them, when faced with challenges in the domain of independent living. Although it was not

(*continued*)

Table 4. Continued.

Author, year (ref)	Country	Study aim or research questions (Quotes)	Knowledge gaps or research needs (Quotes)
			the goal of the study, the current design also did not enable us to structurally differentiate how factors differ between the included types of technology and stages of use. Additionally, many of the phenomena described in our findings are subject to change over time, and research exploring longitudinal mechanisms influencing technology use is required to better understand the dynamics, interplay, and relative importance of factors. More specifically, longitudinal research is needed on how changes in the personal context (i.e., needs, activities, and health status) and the social context (i.e., actors and roles in the social network) affect community-dwelling older adults' attitudes and beliefs with regard to using technology.
Peek et al., 2017 [25]	Netherlands	The current study aimed to understand the origins and consequences of technology acquirement by independent living older adults. We did this by exploring: (1) how and why technologies are acquired by independent-living older adults; and (2) the implications of the ways in which independent-living older adults acquire technologies	Looking at our model and findings, there are several areas that could benefit from further research. First, our design focussed on exploring why independent-living seniors acquired devices, and not on why they did not. Further research is necessary to understand the context and mechanisms of acquirement processes that are not started, or are aborted. This type of research may also lead to insights on mechanisms that impede acquirement, and mechanisms that limit the enabling mechanisms that we have described. Second, the current study solely describes older adults' perceptions of their status quo and developments and mechanisms that led to acquirement. Our model could be expanded by also integrating the perspectives of older adults' spouses and relatives, as well as care organizations they interact with. It would be interesting to integrate their perspectives on the older adults' status quo, their views on what mechanisms influence acquirement, and their motivations for providing resources for acquirement. This also would entail collecting more information on the size and nature of older adults' social networks. As others have pointed out, successful ageing in place is socially and collaboratively accomplished. Third, our model could benefit from better specifying the role of affect in technology acquirement processes. While emotions were part of the stories told by participants, we feel that using qualitative methods may not be the best way to capture their precise role. Quantitative research, for example by employing scales developed by Bagozzi, may shed light on emotional involvement in the adoption process, by measuring anticipated and anticipatory emotions. Based on our findings, we believe that understanding the role of emotions may be particularly important in novel (unfamiliar) types of acquirement. Finally, our participants' views and contexts, as well as their acquirements of technology are likely to be influenced by cultural aspects and the organization of the local and national health care system. Studies in other regions and countries are necessary to determine if our results can be generalized.
Ramsey et al., 2016 [36]	USA	To identify barriers to use of technology for behavioural health care from the perspective of care decision makers at community behavioural health organizations.	While this study was focussed on key care decision makers, future research efforts should strive to capture the perspectives of a more diverse group of health care workers. Regarding organization-level differences, more frequent endorsement of barriers in lower budget and smaller agencies (i.e., structural characteristics of the inner organizational setting) may reflect the multiple challenges in technology-related implementation faced by agencies with relatively few resources. Special efforts should be made with lower resource agencies to employ strategies responsive to multiple implementation barriers and to perform demonstration trials in these organizations as a potential means for increasing

(*continued*)

Table 4. Continued.

Author, year (ref)	Country	Study aim or research questions (Quotes)	Knowledge gaps or research needs (Quotes)
			subsequent confidence, skills, and desire for more sustained uses of technology
Riikonen et al., 2013 [29]	Finland	In this study, answers are sought to the following question: which factors facilitate the use of technology in the daily life of home-living people with dementia?	In future research, investigators should explore how the overall service process could be developed so that we could promptly provide up-to-date information of the latest assistive technologies and their application, flexibly responding to the caring staff's, families' and affected persons' needs for guidance.
Van Heek et al., 2017 [27]	Germany	1. How do the participants evaluate a holistic AAL system (see 3.3) and which perceived benefits and barriers are most important for its acceptance? 2. To which extent do age, experiences with disabilities, and current care needs influence the AAL system's evaluation? 3. Are different benefits and barriers decisive for AAL acceptance depending on diverse user group perspectives?	As the present study was a first approach to compare different user perspectives, it had to be concentrated on the general acceptance of a holistic AAL system and the evaluation of crucial benefits and barriers. In future studies, we will consider other aspects, e.g., relationship between privacy and safety, trade-off between perceived benefits and maybe perceived intrusion of privacy, that have not been taken into account so far. … as this study focussed German participants, it represents a perspective of only one specific country with a specific health care system. For future studies, our approach should be applied in other countries to compare AAL acceptance and future users needs depending on different countries, their specific health care systems, and cultures.
Wang et al., 2019 [31]	USA	The goals of this study were to: (1) involve residents of a local continuing care senior housing community (CCSHC) in conversations about technologies that might facilitate their continued independent living status; (2) assess their privacy attitudes and preferences; and (3) identify whether residents would be interested in co-designing technologies moving forward and if so, how to foster next steps	A next step in this research is to develop a co-design process that incorporates technology education as a component with a goal of increasing "tech literacy." We anticipate this education will facilitate identifying and prioritizing problems that can be addressed with a technological solution that residents help to co-design.
Wangmo et al., 2019 [32]	Switzerland, Germany, Italy	The aim of identifying critical ethical concerns perceived in relation to the use of IATs for the care of older people with dementia	… concerns related to IATs' possible use as a replacement for human care and whether IAT use constitute deception necessitate further research in understanding personal, ideological, social and cultural concerns, and possible mechanisms of addressing them.
Wu et al., 2015 [33]	France	To inform older adults of the existence of different kinds of ICT-related products and to investigate how they perceived this kind of initiative	The participants of the study are mostly older adults who demonstrate "successful ageing". They are globally healthy and independent, enjoy interactions with others, and maintain active engagement in life. Their opinions might not be representative. Frailer older adults with cognitive or physical impairment might have different opinions and attitudes towards ICT-related products. t might be that in frailer older adults, the need to compensate for their impairment would outweigh the need to maintain a certain desired self-image. Future studies could investigate this issue by using a qualitative approach to compare a group of healthy older adults to a group of frail older adults.
Nilsen et al., 2016 [3]	Norway	The aim of this study was to explore resistance to implementation of welfare technology in five municipalities in Norway	In studies as the one at hand, many factors influence the context. In order to reduce complexity, we have omitted several factors. Central and important stakeholders like the patients and next of kin have not been included in the study. This is because we wanted to focus on the employees, but at the same time, we recognize the patient and his/her family as the real end user of the welfare technology. Focus on the patient and families will need to be included in future studies.
Okonji and Ogwezzy, 2019 [30]	Nigeria	This study explored barriers to adoption of assistive technologies among visually impaired people in five southern states in Nigeria	Future studies should investigate how the type of vision impairment and visual acuity could determine awareness and decisions to adopt computer assistive technologies. In addition, the list of possible barriers is not exhaustive of the spectrum of barriers to adoption of ACTs. Future research should explore a broader list of barriers. While our sample may be a true reflection of prevalence of vision impairment among the age-groups (i.e., older people are more affected with vision impairment than younger adults because vision impairment is associated with ageing), there are possibilities that the results are

(*continued*)

Table 4. Continued.

Author, year (ref)	Country	Study aim or research questions (Quotes)	Knowledge gaps or research needs (Quotes)
			skewed to reflect more of the responses of the older adults' age group than the younger. Future research may focus on covering larger and equally sampled age groups.
Olsson et al., 2013 [26]	Sweden	The aim of the present intervention study was to describe and explore the use and experiences of using a passive positioning alarm, over time, in daily life among persons with dementia and their spouses	… future research is needed to explore the experiences of persons with dementia in later stages of the disease. Persons with severe dementia may experience deterioration of their verbal communication skills, which is why a longitudinal observation study should be considered.
Olsson et al., 2016 [38]	Sweden	To describe perceptions of a passive positioning alarm (PPA) among persons with mild dementia	Future development of advanced technology for PwMDs [people with mild dementia] should include potential users' perceptions and experiences of the design prior to implementation. If this is done, we believe interventions will truly live up to their potential to support both PwMDs and their relatives.
Peek et al., 2016 [23]	Netherlands	To explore which factors influence the level of use of various types of technology by older adults who are ageing in place and to describe these factors in a comprehensive model	Additionally, more research is needed to better understand how older adults evaluate and decide between the various (technological) options that are available to them, when faced with challenges in the domain of independent living. Although it was not the goal of the study, the current design also did not enable us to structurally differentiate how factors differ between the included types of technology and stages of use. Additionally, many of the phenomena described in our findings are subject to change over time, and research exploring longitudinal mechanisms influencing technology use is required to better understand the dynamics, interplay, and relative importance of factors. More specifically, longitudinal research is needed on how changes in the personal context (i.e., needs, activities, and health status) and the social context (i.e., actors and roles in the social network) affect community-dwelling older adults' attitudes and beliefs with regard to using technology.
Peek et al., 2017 [25]	Netherlands	The current study aimed to understand the origins and consequences of technology acquirement by independent living older adults. We did this by exploring: (1) how and why technologies are acquired by independent-living older adults; and (2) the implications of the ways in which independent-living older adults acquire technologies	Looking at our model and findings, there are several areas that could benefit from further research. First, our design focussed on exploring why independent-living seniors acquired devices, and not on why they did not. Further research is necessary to understand the context and mechanisms of acquirement processes that are not started, or are aborted. This type of research may also lead to insights on mechanisms that impede acquirement, and mechanisms that limit the enabling mechanisms that we have described. Second, the current study solely describes older adults' perceptions of their status quo and developments and mechanisms that led to acquirement. Our model could be expanded by also integrating the perspectives of older adults' spouses and relatives, as well as care organizations they interact with. It would be interesting to integrate their perspectives on the older adults' status quo, their views on what mechanisms influence acquirement, and their motivations for providing resources for acquirement. This also would entail collecting more information on the size and nature of older adults' social networks. As others have pointed out, successful ageing in place is socially and collaboratively accomplished. Third, our model could benefit from better specifying the role of affect in technology acquirement processes. While emotions were part of the stories told by participants, we feel that using qualitative methods may not be the best way to capture their precise role. Quantitative research, for example by employing scales developed by Bagozzi, may shed light on emotional involvement in the adoption process, by measuring anticipated and anticipatory emotions. Based on our findings, we believe that understanding the role of emotions may be particularly important in novel (unfamiliar) types of acquirement.

(*continued*)

Table 4. Continued.

Author, year (ref)	Country	Study aim or research questions (Quotes)	Knowledge gaps or research needs (Quotes)
			Finally, our participants' views and contexts, as well as their acquirements of technology are likely to be influenced by cultural aspects and the organization of the local and national health care system. Studies in other regions and countries are necessary to determine if our results can be generalized.
Ramsey et al., 2016 [36]	USA	To identify barriers to use of technology for behavioural health care from the perspective of care decision makers at community behavioural health organizations.	While this study was focussed on key care decision makers, future research efforts should strive to capture the perspectives of a more diverse group of health care workers. Regarding organization-level differences, more frequent endorsement of barriers in lower budget and smaller agencies (i.e., structural characteristics of the inner organizational setting) may reflect the multiple challenges in technology-related implementation faced by agencies with relatively few resources. Special efforts should be made with lower resource agencies to employ strategies responsive to multiple implementation barriers and to perform demonstration trials in these organizations as a potential means for increasing subsequent confidence, skills, and desire for more sustained uses of technology
Riikonen et al., 2013 [29]	Finland	In this study, answers are sought to the following question: which factors facilitate the use of technology in the daily life of home-living people with dementia?	In future research, investigators should explore how the overall service process could be developed so that we could promptly provide up-to-date information of the latest assistive technologies and their application, flexibly responding to the caring staff's, families' and affected persons' needs for guidance.
Van Heek et al., 2017 [27]	Germany	1. How do the participants evaluate a holistic AAL system (see 3.3) and which perceived benefits and barriers are most important for its acceptance? 2. To which extent do age, experiences with disabilities, and current care needs influence the AAL system's evaluation? 3. Are different benefits and barriers decisive for AAL acceptance depending on diverse user group perspectives?	As the present study was a first approach to compare different user perspectives, it had to be concentrated on the general acceptance of a holistic AAL system and the evaluation of crucial benefits and barriers. In future studies, we will consider other aspects, e.g., relationship between privacy and safety, trade-off between perceived benefits and maybe perceived intrusion of privacy, that have not been taken into account so far. … as this study focussed German participants, it represents a perspective of only one specific country with a specific health care system. For future studies, our approach should be applied in other countries to compare AAL acceptance and future users needs depending on different countries, their specific health care systems, and cultures.
Wang et al., 2019 [31]	USA	The goals of this study were to: (1) involve residents of a local continuing care senior housing community (CCSHC) in conversations about technologies that might facilitate their continued independent living status; (2) assess their privacy attitudes and preferences; and (3) identify whether residents would be interested in co-designing technologies moving forward and if so, how to foster next steps	A next step in this research is to develop a co-design process that incorporates technology education as a component with a goal of increasing "tech literacy." We anticipate this education will facilitate identifying and prioritizing problems that can be addressed with a technological solution that residents help to co-design.
Wangmo et al., 2019 [32]	Switzerland, Germany, Italy	The aim of identifying critical ethical concerns perceived in relation to the use of IATs for the care of older people with dementia	… concerns related to IATs' possible use as a replacement for human care and whether IAT use constitute deception necessitate further research in understanding personal, ideological, social and cultural concerns, and possible mechanisms of addressing them.
Wu et al., 2015 [33]	France	To inform older adults of the existence of different kinds of ICT-related products and to investigate how they perceived this kind of initiative	The participants of the study are mostly older adults who demonstrate "successful ageing". They are globally healthy and independent, enjoy interactions with others, and maintain active engagement in life. Their opinions might not be representative. Frailer older adults with cognitive or physical impairment might have different opinions and attitudes towards ICT-related products. t might be that in frailer older adults, the need to compensate for their impairment would outweigh the need to maintain a certain desired self-image. Future studies could investigate this issue by using a qualitative approach to compare a group of healthy older adults to a group of frail older adults.

adoption among people with visual impairments and how visual impairment could determine awareness and decisions to adopt assistive technology [30]. Knowledge about longitudinal mechanisms that influence the use of technology is also called for, as it will help in understanding the dynamics, interplay and relative importance of relevant factors [23]. In particular, studies of the experiences of persons with dementia in later stages of the disease have been requested [26].

Attitudes, beliefs and context

The literature has issued a general call for studies to address barriers related to person and context [21,28]. More specific research needs include comparisons of opinions and attitudes among healthy and frail older adults [33]. There are also needs for studies to understand personal, ideological, social and cultural concerns and possible mechanisms of addressing them [32]. Furthermore, studies to explore how changes in personal and social contexts affect older adults' attitudes and beliefs with respect to using technology while ageing in place have been called for [23]. This includes a need for knowledge about the size and nature of older adults' social networks [33].

The literature has reported a need to address gaps in knowledge about ethical implications for individuals and societies [20]. In particular, studies on ethical principles of autonomy and rights to self-determination [21], and the relationship between safety and perceived intrusion of privacy [27] have been requested. Moreover, studies of the role of culture, reflecting beliefs and being an integral part of the context, have been requested, including the role of culture in design, marketing, acquirement and adoption [21,25].

Caregiver perspectives

The reviewed articles expressed a need for studies on caregiver perspectives [21], including those of patients and families as end users of welfare technology [3] and different healthcare workers [36]. Specific calls have been made for comparisons of care tasks and task pressure among different groups of informal caregivers, and research on how contextual factors, such as culture, age and personal innovativeness, affect the perception of welfare technology among informal caregivers [24]. There is also a reported need for integrating perspectives of spouses and relatives of older adults, as well as care organizations they interact with, on the influences and motivation for providing resources for acquiring technology [25]. Furthermore, among informal caregivers, there is a gap in knowledge about the male perspective [24].

Services

Knowledge gaps have been identified concerning the meaning of drivers and barriers when technologies are integrated in everyday care practices [24]. There is also a reported knowledge gap about strategies that are responsive to multiple implementation barriers faced by lower resource agencies [36]. Another area where more knowledge has been called for is how the overall service process could be developed to provide stakeholders with up-to-date information on the latest technologies and their applications [29]. Welfare technology costs are also an area that requires exploration [21], particularly the multiple challenges in technology-related implementation faced by small-and low-budget agencies [36].

Technology

Besides the general need for research about technology-related barriers [28], there is a need for studies on features, functionality, human factors engineering and costs [21]. Moreover, research is needed about the development of technical solutions that are sophisticated, safe, personalizable, non-stigmatizing [37] and based on potential users' perceptions and experiences of the design prior to implementation [38]. However, before embarking on developing new solutions, it is necessary to identify and priorities problems that can be addressed [31]. Finally, reviews of socio-technical challenges in relation to the individual, organization and community levels have been called for [34].

Discussion

The results of the analysis of the data extracted from the 21 empirical research articles included in the review form a comprehensive understanding that knowledge gaps and research needs exist in areas important for improved implementation of welfare technology for older people, people with disabilities and informal caregivers. Two categories of knowledge gaps and research needs emerge, namely, research designs and populations, and focus of research. Needs were reported for comparative studies, demonstration trials, studies employing longitudinal designs, systematic reviews, and developing co-design processes. Furthermore, the literature called for studies within and across countries that involve older people and people with disabilities, informal and formal caregivers, other professions, and care organizations. Sometimes, populations were specified together with particular research designs or methods. The reported knowledge gaps cover a broad spectrum of issues related to the implementation of welfare technology. They translate into the need for studies on barriers and facilitators of acquirement, adoption, and acceptance of welfare technology as well as studies of the roles of attitudes, beliefs, and context for the implementation of welfare technology, including ethics and culture. Moreover, the need was expressed for studies of informal and formal caregiver perspectives on welfare technology, services for the provision of welfare technology and welfare technology itself. This variety of calls for research in implementation of welfare technology depicture a very broad list of different perspectives illustrating an emerging interdisciplinary field of research. Aaen [39] reported similar findings in a review exploring competing concerns in managing the critical transition from small-scale welfare technology inventions to large-scale implementation. The findings were presented in eight themes using a system theory structure divided into user level, organizational level, market level and policy level. Similarly, Cunningham and O'Reilly [40] call for a social system theory approach in a research agenda about technology transition, which implementation of welfare technology is part of. They propose to evaluate countries and make cross-country comparisons at macro-level, followed by focussing on technology transition at meso level. And, at micro level, they suggest research having individual perspectives focussing behaviours and actions.

This review identified significant knowledge gaps and needs for research in areas important for improved and large-scale implementation of welfare technology, which has been slow and faced numerous barriers [1–9]. Addressing these gaps and needs will generate knowledge that facilitates the successful implementation of welfare technology. Particularly, interdisciplinary research that generates high-quality generalizable evidence is much needed in this field, including longitudinal studies of the effects of using welfare technology for individuals, informal and formal caregivers, organizations, and society.

The findings of this review resonate with a global priority research agenda for improving access to assistive technology,

which includes welfare technology. The agenda has identified five thematic areas in need of urgent attention. Among others, the thematic areas cover different technical solutions and their impacts, implementation in terms of policies, systems, models and practices and human resource development, all of which are relevant to welfare technology and its implementation [41].

This study was a state- of-the art review, a method having both strengths and limitations. The aim with the method was to address a current matter. The method offers new perspectives and points out areas for further research and research opportunities for contemporary knowledge gaps and needed research [14]. The method made it possible to present an overview of current knowledge gaps and research needs as reported in primary scientific studies addressing the implementation of welfare technology for older people, people with disabilities and informal caregivers. Considering that welfare technology and its implementation is an emerging interdisciplinary field, it was expected that the database searches would yield a limited number of relevant research studies. Although the search strategy resulted in many hits, many studies were excluded due to the use of concepts not fitting the present study's inclusion criteria and the conceptualization of welfare technology. A critical phase in a review study, is always the search process as the vital part of the study. In the present study, by using the concept welfare technology (applied as the phrase "welfare technology), made the search process problematic. As the concept welfare technology is mainly used within the Nordic countries, comparable or complementary concepts were included in the search process. Most results were retrieved from the more technology-specific terms, such as "alarm" or "smart home*," while articles containing the phrase "welfare technology" were found only in studies originating from the Nordic countries. To prevent too many hits for this review, generic terms, such as "assistive technology", were not used in the search strategy. However, we did use more specific terms, including "digital assistive" and "intelligent assistive". By using more collective concepts, there is a risk that relevant studies using specific terms have been missed.

Conclusion

This review offers an overview of current knowledge gaps and research needs regarding the implementation of welfare technology for older people, people with disabilities and informal caregivers. There are considerable knowledge gaps and research needs related not only to the focus of research but also to research designs and study populations. Addressing these gaps and needs is important for improving the implementation of welfare technology. A research agenda with interdisciplinary research perspectives, and the application of micro, meso and macro levels with social system theory approaches, appear promising when responding to identified knowledge gaps and research needs.

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